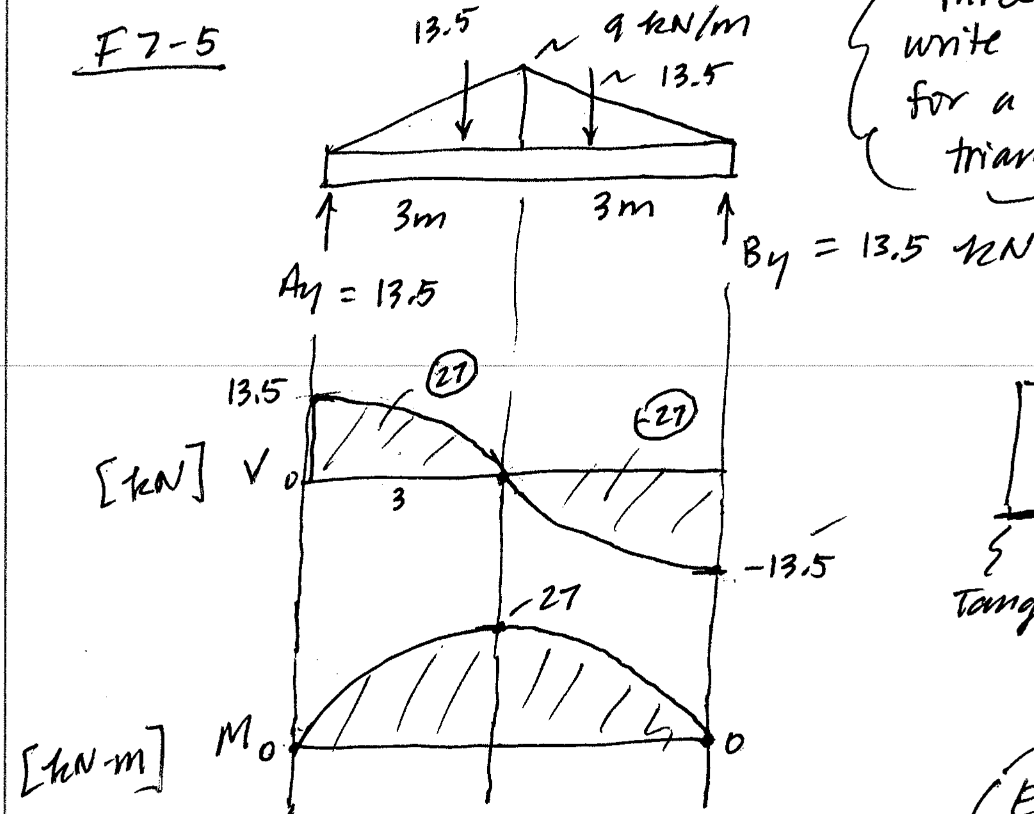
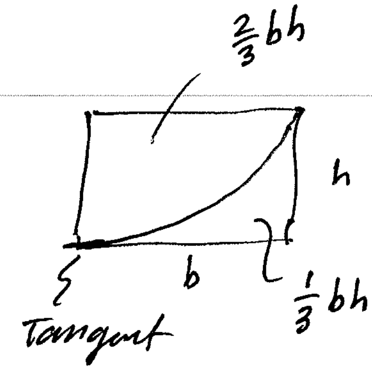
F7-5

Three ways to write V and M Eqns for a descending triangular load.

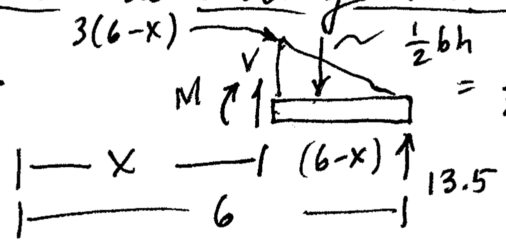




① Work as ascending triangle

Easiest Way

$3 \le x \le 6$



$= \frac{3}{2}(6-x)^2$

$\frac{w}{L} = \frac{9}{3} = 3$

Take FBD of RHS of beam to make it ascending!

$+\uparrow \quad V + 13.5 - \frac{3}{2}(6-x)^2$

$$\boxed{V = \frac{3}{2}(6-x)^2 - 13.5}$$

$+\circlearrowleft \sum M_{cut} = 0; \quad M - 13.5(6-x) + \frac{3}{2}(6-x)^2 \frac{(6-x)}{3} = 0$

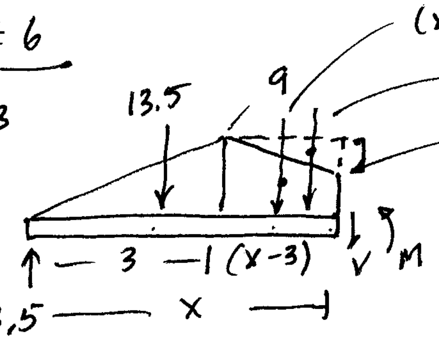
$$\boxed{M = 13.5(6-x) - \frac{1}{2}(6-x)^3}$$

x	V	M
3	0	27
6	-13.5	0

$\boxed{\text{Model descending load as } \square - \triangle}$

(2) $3 \le x \le 6$

$\frac{w}{L} = \frac{9}{3} = 3$



$= \frac{3}{2}(x-3)^2$

Subtract the triangle

$+\uparrow \quad \cancel{13.5} - \cancel{13.5} - \underbrace{9(x-3) + \frac{3}{2}(x-3)^2}_{-\left[9(x-3) - \frac{3}{2}(x-3)^2\right]} - V = 0$

$$\boxed{V = -9(x-3) + \frac{3}{2}(x-3)^2}$$

$\overset{+}{\circlearrowleft} \Sigma M_{cut} = 0; \quad M - 13.5x + 13.5(x-2) + 9(x-3)\frac{(x-3)}{2} - \frac{3}{2}(x-3)^2\left(\frac{x-3}{3}\right) = 0$

$$\boxed{M = 13.5x - 13.5(x-2) - 4.5(x-3)^2 + \frac{1}{2}(x-3)^3}$$

$x=3$

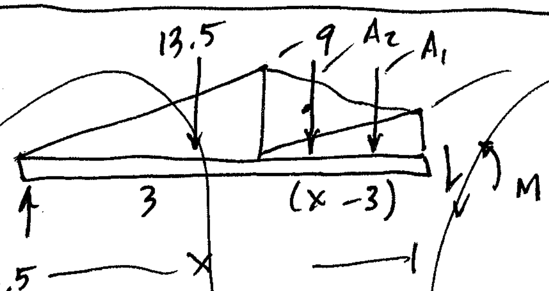
40.5
− 13.5
+27

$x=6$

81
− 54
− 40.5
+ 13.5
0

x	V	M
3	0	27
6	−13.5	0

(3)



$\frac{w}{L}(x-3) = 3(x-3)$

$A_1 = \frac{1}{2}bh$
$= \frac{1}{2}(x-3)3(x-3)$
$= \frac{3}{2}(x-3)^2$

$A_2 = \frac{1}{2}bh = \frac{1}{2}9(x-3) = 4.5(x-3)$

$+\uparrow \quad \cancel{13.5} - \cancel{13.5} - 4.5(x-3) - \frac{3}{2}(x-3)^2 - V = 0$

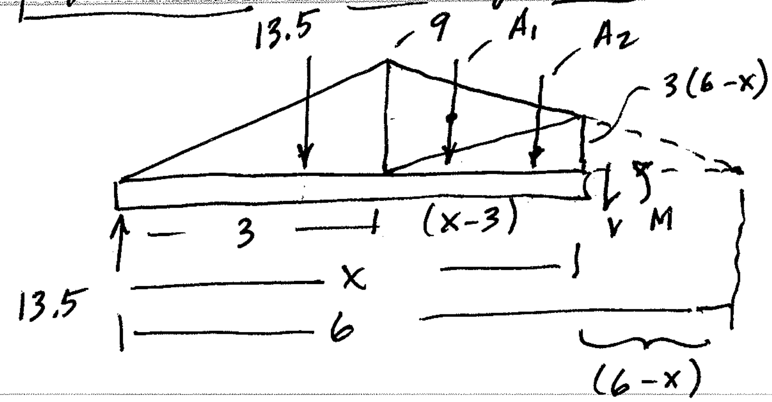
$$\boxed{V = -4.5(x-3) - \frac{3}{2}(x-3)^2}$$

x	V
3	0
6	

−13.5
−13.5

Model as two triangles

(3)



$\frac{w}{L} = \frac{9}{3} = 3$

$A_1 = \frac{1}{2}bh$
$= \frac{1}{2}(9)(x-3)$
$= 4.5(x-3)$

$\underline{3 \le x \le 6}$

$A_2 = \frac{1}{2}bh = \frac{1}{2}(x-3)(3(6-x))$
$= \frac{3}{2}(x-3)(6-x)$

$$+\downarrow \quad V - \cancel{13.5} + \cancel{13.5} + 4.5(x-3) + \frac{3}{2}(x-3)(6-x) = 0$$

$$\boxed{V = -4.5(x-3) - \frac{3}{2}(x-3)(6-x)}$$

x	V
3	0
6	-13.5

Checks!

$$\circlearrowleft + \ \Sigma M_{cut} = 0; \quad M - 13.5x + 13.5(x-2) + 4.5(x-3)\frac{2}{3}(x-3)$$
$$+ \frac{3}{2}(x-3)(6-x)\frac{1}{3}(x-3) = 0$$

$$\boxed{M = 13.5x - 13.5(x-2) - 3(x-3)^2 - \frac{1}{2}(x-3)^2(6-x)}$$

x	M
3	27
6	0

Checks!